

Pros and cons of external swabbing of amphibians for genetic analyses

Antonia S. Müller · Patrick P. Lenhardt · Kathrin Theissinger

Received: 5 December 2012 / Revised: 11 June 2013 / Accepted: 16 June 2013 / Published online: 2 July 2013
© Springer-Verlag Berlin Heidelberg 2013

Abstract Non-invasive DNA sampling is an important tool in amphibian conservation. Buccal swabs are nowadays replacing the wounding toe-clipping method. Skin and cloaca swabbing are even less invasive and easier to handle than buccal swabbing, but could result in contaminations of genetic material. Therefore, we test if external skin and cloaca swabs are as reliable as buccal swabs for genetic analysis of amphibians. We analysed eight microsatellite loci for the common frog (*Rana temporaria*, Linnaeus 1758) and compared genotyping results for buccal, skin and cloaca swabs regarding allelic dropouts and false alleles. Furthermore, we compared two DNA extraction methods regarding efficiency and cost. DNA quality and quantity (amplification success, genotyping error rate, in nanogram per microlitre) were comparable among DNA sources and extraction methods. However, skin and cloaca samples exhibited high degrees of contamination with foreign individuals, which was due to sample collection during mating season. Here, we established a simple low budget procedure to receive DNA of amphibians avoiding stressful buccal swabbing or harmful toe clipping. However, the possibility of contaminations of external swabs has to be considered.

Keywords Common frog · DNA extraction · Reliable genotypes · Microsatellites · DNA contamination

Introduction

The global amphibian decline is a well-described problem. Stuart et al. (2004) showed that 43.2 % of amphibian species

are experiencing some form of population decrease. Apart from climate change, over-exploitation of land and introduction of alien species, the main reasons for this decline are habitat loss and fragmentation (Stuart et al. 2004; Beebe and Griffiths 2005). For the protection of amphibian populations, information about their genetic structure is necessary to answer questions concerning population size, genetic diversity and gene flow among populations (Frankham et al. 2002). Toe clipping is a common method to collect genetic material of amphibians. However, it often leads to infections and reduction in mobility (McCarthy and Parris 2004). To avoid harmful injuries, the buccal swab method was established (Poschadel and Möller 2004), which is easy to perform, as cheap as other sampling procedures (Pidancier et al. 2003) and resulting in good DNA quality (Broquet et al. 2007). Moreover, storage and transport of swabs is uncomplicated (Poschadel and Möller 2004), and it is easier to get sampling permission for endangered species (Pidancier et al. 2003). In comparison to toe clipping, buccal swabbing is minimally invasive (Broquet et al. 2007). Nevertheless, opening the mouth of amphibians can be challenging and torturous, especially in small amphibians (e.g. newts). Prunier et al. (2012) already demonstrated the reliability of skin swabs compared to buccal swabs. However, contaminations of external swabs with DNA of foreign individuals are possible especially during mating season or when target species get in close contact during the sampling procedure. Therefore, we tested the reliability of skin and cloaca swabs compared to buccal swabs for genetic analysis of amphibians. Moreover, we compared the effectiveness of two differing DNA extraction methods for swab samples.

Materials and methods

We used as model organism the common frog (*Rana temporaria*) due to an easy handling and microsatellite

Communicated by C. Gortázar

A. S. Müller (✉) · P. P. Lenhardt · K. Theissinger
Institute for Environmental Science, University of Koblenz
Landau, 76829 Landau, Germany
e-mail: muellertoniam@gmail.com

marker availability. Samples were collected in Germany, Rhineland-Palatinate, near Ramstein-Miesenbach (49.27°N, 7.34°E) during mating season in March 2012. From 20 frogs, three swabs were taken each from buccal mucosa, skin and cloaca ($N=60$). We used sterile cotton swabs with a wooden handle and separate plastic cover (COPAN, Brescia, Italy). To take the buccal samples, the mouth was opened carefully with a flat plastic tip, and the swab was rolled over the inside of the cheek. For skin sampling, the cotton swab was rolled thoroughly over back, stomach and legs. For cloaca sample, the swab was rolled over the external cloaca. Samples were kept in a cooler box and stored at -23°C .

Two DNA extraction methods were tested. Extractions of 44 samples (14 buccal, 15 skin, 15 cloaca) were conducted according to high salt DNA extraction protocol after Aljanabi and Martinez (1997) (ca. 0.10 €/sample). DNA of 16 samples (six buccal, five skin, five cloaca) was isolated using the Macherey Nagel NucleoSpin Tissue Kit (Düren, Germany; 2.18 €/sample) following the manufacturer's protocol. Nucleic acid concentrations were measured with NanoDrop ND-1000 Spectrophotometer (Peqlab, Erlangen, Germany).

We analysed eight variable microsatellite loci (Table 1; Matsuba and Merliä 2009) and amplified the fragments in two multiplex PCRs by using the QIAGEN Multiplex PCR Kit (Hilden, Germany). Primers were fluorescently labelled, and 10 pM of primer working solutions were mixed in two multiplex reactions in relation of 1 (Cy5):2 (Cy5.5):4 (Cy5.7). The PCR was conducted in 5 μL of final volume containing 2.5 μL QIAGEN Master Mix, 1.2 μL QIAGEN Solution, 0.3 μL Primer Mix and 1 μL DNA. PCR was conducted with Primus 96 Cyclet (Peqlab Biotechnologie GmbH, Erlangen, Germany) with the following thermal profile: initial denaturation at 95°C for 15 min, followed by 30 cycles of 30 s at 94°C , 1.5 min at 55°C and 1 min at 72°C . A terminal elongation step was applied for 30 min at 60°C . Amplification products were run on the CEQ 8000 Sequencer from Beckman Coulter (Krefeld, Germany). Buccal samples were repeated in a multi-tube approach (Taberlet et al. 1996) at least three times to identify real genotypes. Alleles had to be identical across at least two repetitions for heterozygotes and three repetitions for homozygotes (Hansen et al. 2008; Kolodziej et al. 2013). Skin and cloaca samples were repeated up to three times only in case of low amplification success or technical problems.

Fragments were analysed with the software GeneMarker 1.95 (SoftGenetics, State College, Pennsylvania). The genotypes of the buccal, skin and cloaca swabs of the same individual were compared for consistency, and allelic drop-outs (ADO) as well as false alleles (FA) were counted. Genotyping error rates (GER) were calculated as the sum of ADO and FA for each buccal, skin and cloaca swabs per locus and across all loci. If skin or cloaca genotypes

exhibited differing or additional alleles compared to the buccal genotype on at least two loci, we assigned this as a contamination with genetic material of additional individuals, which is also demonstrated by the high number of FAs found in skin and cloaca swabs which seem strongly correlated (Fig. 1).

Expected (H_E) and observed (H_O) heterozygosity were calculated with GENETIX 4.04 (<http://kimura.univmontp2.fr/genetix/>) based on buccal swab results. GER for extraction methods was calculated with buccal swab results as well. Percentage of genotyping success for each swabbing method was calculated over all loci and all repetitions. The raw data table of microsatellite alleles has been uploaded at the dryad database (<http://dx.doi.org/10.5061/dryad.4qb53>) and can be viewed under link (will be provided after acceptance for publication; excel spread sheet attached).

Results and discussion

DNA quantity was comparable among both extraction methods. For the high salt extraction procedure, DNA concentrations ranged from 6.0 to 403.2 ng/ μL (median=47.6 ng/ μL ; lower quartile=35.7 ng/ μL ; upper quartile=76.3 ng/ μL). For the commercial kit, DNA concentration ranged from 18.0 to 303.3 ng/ μL (median=66.7 ng/ μL ; lower quartile=58.9 ng/ μL ; upper quartile=123.0 ng/ μL). The high salt DNA extraction resulted in a GER of 0.3 %; for the commercial kit extraction, the GER was 0.9 % (Table 1). Thus, high salt extraction method is a reliable and cheap alternative to commercial kits. Nucleic acid concentration in DNA extracts of the different swabs ranged from 12.0 to 242.2 ng/ μL (median=44.0; upper quartile=63.5; lower quartile=32.9) for buccal swabs; from 8.0 to 207.0 ng/ μL (median=51.7; upper quartile=82.5; lower quartile=26.4) for skin swabs; and from 8.1 to 403.0 ng/ μL (median=56.5; upper quartile=82.3; lower quartile=44.0) for cloaca swabs. This shows that extracted DNA quantity is comparable among the swab sources.

The genotyping success (GS), i.e. the amplification of each locus, was 71.8 % for buccal swabs, 41.4 % for skin swabs and 51.9 % for buccal swabs over all loci, which was mainly due to the non-amplification of one locus (BFG_099). Excluding BFG_099, the GS across the seven remaining loci resulted in 85.1, 62.1 and 72.2 %, respectively, which is well in the typical range for microsatellite studies using forensic samples (GS 20–70 %; Kolodziej et al. 2011). For the buccal swabs, the multi-tube approach resulted in a GER of only 0.4 % due to ADOs (Table 1). In comparison to other studies using forensic samples, this is a very low GER. Broquet (2004) showed an ADO rate of 11.3 % for faeces and 18.7 % for hair; in the study of Kolodziej et al. (2013) with faeces of wild boar, a GER of

Table 1 Summary data table of eight microsatellite loci. Given are expected (H_E) and observed (H_O) heterozygosity, allelic dropout rate (ADO), false allele rate (FA) and total genotyping error rate

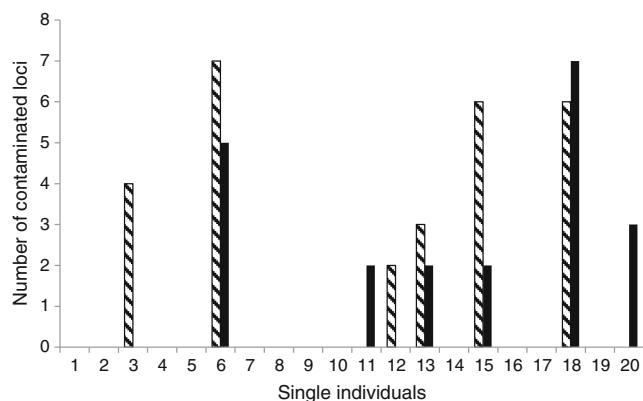
(GER=ADO+FA) for buccal, skin and cloaca swabs per locus and over all loci. For buccal swabs, total GER was calculated for both extraction methods additionally

Locus	BFG066	BFG151	BFG130	BFG082	BFG099	BFG160	BFG145	BFG129	Over all loci
Repeat motif	AAG	GAAA	TCTT	TATC	ACTC	TCTA	TCTA	CTAT	
Allele range	143–182	302–394	327–319	207–307	143–195	201–253	154–210	264–628	
Multiplex set	A	A	A	B	B	B	B	B	
H_E/H_O	0.78/0.60	0.87/0.90	0.92/1.00	0.88/0.37	0.78/0.58	0.88/0.79	0.90/0.90	0.88/0.65	0.86/0.72
Buccal swabs									
Number ^a	60	60	60	55	45	56	58	57	
ADO rate (%)	0.0	0.0	1.7	0.0	0.0	1.8	0.0	0.0	0.4
FA rate (%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total GER (%)	0.0	0.0	1.7	0.0	0.0	1.8	0.0	0.0	0.4
GS (%)	100.0	100.0	91.5	90.8	72.4	92.7	97.7	96.6	71.8
Total GER (%) for extraction methods									
HSE	0.0	0.0	2.4	0.0	0.0	0.0	0.0	0.0	0.3
TK	0.0	0.0	0.0	0.0	0.0	7.1	0.0	0.0	0.9
Skin swabs									
Number ^a	29	27	24	26	16	21	26	29	
ADO rate (%)	0.0	11.1	0.0	0.0	0.0	0.0	0.0	3.4	1.8
FA rate (%)	20.7	7.4	16.7	15.4	0.0	9.5	3.8	10.3	10.5
Total GER (%)	20.7	18.5	16.7	15.4	0.0	9.5	3.8	13.7	12.3
GS (%)	100.0	100.0	91.5	90.8	72.4	92.0	97.7	97.7	41.4
Cloaca swabs									
Number ^a	28	28	25	22	14	21	24	25	
ADO rate (%)	0.0	3.6	0.0	9.1	0.0	0.0	0.0	0.0	1.6
FA rate (%)	25.0	28.6	19.2	4.5	0.0	14.3	8.3	0.0	12.5
Total GER (%)	25.0	32.2	19.2	13.6	0.0	14.3	8.3	0.0	14.1
GS (%)	100.0	100.0	96.4	76.9	57.7	73.1	96.2	100.0	51.9

GS percentage genotyping success, HSE high salt extraction, TK Macherey Nagel NucleoSpin Tissue Kit

^a Number of successful amplifications

4.3 % occurred. Although faeces and hair samples come along with more potential inhibition problems than buccal swabs,

**Fig. 1** Number of contaminated loci per individual. Black bars, cloaca; striped bars, skin

this comparison demonstrates that laboratory and scoring procedures were highly reliable in our study. Compared to the buccal swabs, the skin and cloaca swabs exhibited high GERs of 12.3 and 14.1 %, respectively, mainly due to FAs (Table 1). Identical genotypes for all three types of swabs were observed at only 12 of the 20 analysed individuals (60 %). Eight individuals exhibited divergent genotypes at one to seven loci for skin and/or cloaca swabs when compared to the corresponding buccal swab (Fig. 1). Our data set partly exhibited the consistency of FAs across skin and cloaca swabs at some loci ($N=7$), indicating that the high GERs of skin and cloaca swabs was not due to technical problems. The fact that skin and cloaca swabs did not show the same FAs at all instances might be due to the even lower DNA traces of the foreign individuals (e.g. sexual partner), which amplify inconsistently by chance. Moreover, we show the occurrence of four-allele combinations at some loci of skin and cloaca swabs ($N=3$) (<http://dx.doi.org/10.5061/drayad.4qb53>), indicating that the

high GERs could be attributed to contaminations of target DNA with DNA of a different individual.

This possible contamination could be due to sampling during mating season, when *R. temporaria* is most easily accessible. During the mating procedure, one or several males grasp the females with their forelegs, forming a big lump of frogs. Therefore, the external swabbing of amphibians during mating season, especially for explosive breeders like the common frog or the common toad, could result in high contamination rates. Consequently, skin and cloaca swabs should be applied outside the mating season only. However, this may increase the sampling effort compared to the high and centralised abundance at breeding ponds during the mating season. Cleaning the skin of frogs or toads with water before swabbing may improve the quality of swabs though, also for amphibians caught at fences in buckets. Nevertheless, for newts, e.g. those which are typically caught individually in bottles, skin and cloaca swabs might be feasible.

Beja-Pereira et al. (2009) reviewed that non-invasive genetic sampling is nowadays common in wildlife studies and that buccal, skin and cloaca swabs of amphibians and reptiles provide reliable DNA sources for microsatellite analyses. Prunier et al. (2012) also showed that skin swabs are a good alternative to buccal swabs especially for small and vulnerable amphibians. Also for reptiles, cloaca swabs are a reliable alternative to blood sampling and toe clipping (Miller 2006; Lanci et al. 2012). With this study, we agree that skin, but also cloaca, swabs are easier to conduct, come with a lower risk of injuries for the amphibians and produce comparably good-quality DNA for population genetic analyses. However, we showed that skin and cloaca swabs are not a reliable method of DNA sampling for all amphibian species regarding the respective collection practices. DNA contaminations could lead to distortion of genotyping results what eminently inhibits further analysis. To totally omit the possibility of contaminations with foreign individuals, we recommend that external swabbing should only be conducted if contact of individuals can be excluded.

Acknowledgments Thanks to Jürgen Ott for help with organising the field work, to Maria Piscione and Stefan Müller for field assistance, and to Barbara Thronicke for support in the laboratory. This project was authorised by the “Struktur und Genehmigungsdirektion Süd” (ref. number 42/553-254).

References

- Aljanabi S, Martinez I (1997) Universal and rapid salt-extraction of high quality genomic DNA for PCR-based techniques. *Nuc Acids Res* 25(22):4692–4693
- Beebee TJC, Griffiths RA (2005) The amphibian decline crisis: a watershed of conservation biology? *Biol Conserv* 125:271–285
- Beja-Pereira A, Oliveira R, Alves PC, Schwartz MK, Luikart G (2009) Advancing ecological understandings through technical transformations in noninvasive genetics. *Mol Ecol Res* 9:1279–1301
- Broquet T, Berset-Braendli L, Emaresi G, Fumagalli L (2007) Buccal swabs allow efficient and reliable microsatellite genotyping in amphibians. *Conserv Genet* 8:509–5011
- Broquet T, Petit E (2004) Quantifying genotyping errors in noninvasive population genetics. *Mol Ecol* 13:3601–3608
- Frankham R, Ballou JD, Briscoe DA (2002) Introduction to conservation genetics. Cambridge University Press, Cambridge
- Hansen H, Ben-David M, McDonald D (2008) Effects of genotyping protocols on success and errors in identifying individual river otters (*Lontra canadensis*) from their faeces. *Mol Ecol Res* 8:282–289
- Kolodziej K, Theissinger K, Brün J, Schulz HK, Schulz R (2011) Determination of minimum number of microsatellite markers for individual genotyping in wild boar (*Sus scrofa*) using a test with close relatives. *Eur J Wildl Res* 58:621–628
- Kolodziej K, Schulz HK, Theissinger K, Ebert C, Hohmann U, Schulz R (2013) Comparison of established methods for quantifying genotyping error rates in wildlife forensics. *Conserv Genet Resour* 5:287–292
- Lanci AKJ, Roden SE, Bowman A, LaCasella E, Fray A, Dutton PH (2012) Evaluating buccal and cloaca swabs for ease of collection and use in genetic analyses of marine turtles. *Chelonian Conserv Biol* 11:144–148
- Matsuba C, Merliä J (2009) Isolation and characterization of 145 polymorphic microsatellite loci for the common frog (*Rana temporaria*). *Mol Ecol Res* 9:555–562
- McCarthy MA, Parris KM (2004) Clarifying the effect of toe clipping on frogs with Bayesian statistics. *J App Ecol* 41:780–786
- Miller HC (2006) Cloaca and buccal swabs are a reliable source of DNA for microsatellite genotyping of reptiles. *Conserv Genet* 7:1001–1003
- Pidancier N, Miquel C, Miaude C (2003) Buccal swabs are a non-destructive tissue sampling method for DNA analysis in amphibians. *Herpetol J* 13:175–178
- Poschadel J, Möller D (2004) A versatile field method for tissue sampling on small reptiles and amphibians, applied to pond turtles, newts, frogs and toads. *Conserv Genet* 00:1–3
- Prunier J, Kaufmann B, Grolet O, Picard D, Pompanon F, Joly P (2012) Skin swabbing as a new efficient DNA sampling technique in amphibians, and 14 new microsatellite markers in the alpine newt (*Ichthyosaura alpestris*). *Mol Ecol Res* 12(3):524–531
- Stuart S, Chanson J, Cox N, Young B, Rodrigues A, Fischman D, Waller R (2004) Status and trends of amphibian declines and extinctions worldwide. *Science* 306:1783–1786
- Taberlet P, Griffin S, Goossens B, Questiau S, Manceau V, Escaravage N, Waits L, Bouvet J (1996) Reliable genotyping of samples with very low DNA quantities using PCR. *Nuc Acids Res* 24:3189–3194